

Digital Material Passport

ID DMP-2024-ARR-001 - Version 0.1.0
Issue Date 2024-12-15 - Certificate Type EN 10204 3.1

Manufacturer	Customer	Business Transaction	
Advanced Steel Technologies GmbH Industriestraße 15 52070 Aachen DE quality@advanced-steel.de	Precision Engineering Ltd. Manufacturing Park 42 Birmingham B1 1AA GB procurement@precision-eng.co.uk	Order	PO-2024-7890
		Delivery	DN-2024-3456 - 500 kg

Product

Product Name	Hardenability Test Steel	Shape	RoundBar - D 100 mm - Length 1000 mm
Batch ID	HTB-2024-045		

Chemical Analysis

Heat Number H-2024-078 - Sample Location Ladle

Symbol	C	Mn	Si	P	S	Cr	Ni
Unit	%	%	%	%	%	%	%
Min	0.42	0.5					
Max	0.5	0.8	0.4	0.03	0.025	0.25	0.25
Actual	0.45	0.65	0.25	0.015	0.008	0.12	0.08

Mechanical Properties

Property	Symbol	Actual	Minimum	Maximum	Method	Status
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Jominy Hardenability Test					ISO 377	✓
Quenched from 850°C						

Distance from - quenched end (mm)	1.5	3	5	7	9	11	15	20	25	30
Value [HRC]	45	43	40	36	33	31	27	25	28	27
Min	42	39	35	32	29	26	22	20		
Max	47	46	44	41	39	37	33	31	<= 30	<= 29
Status	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass

Coating Thickness Profile	ISO 2178	✓
Magnetic induction method		

Position along length (mm)	0	50	100
Value [µm]	125	115	108
Min	>= 120	>= 110	>= 100
Max	<= 130	<= 125	<= 120
Status	Pass	Pass	Pass

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with EN 10083-1 and ISO 377. All hardenability requirements have been met.

Validated by **Dr. Maria Schmidt**, Metallurgical Engineer,
Quality Control Laboratory - 2024-12-15